FILTER DESIGN TEST CASE

Notch Filter - Band 3 (1800 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Notch Filter filter targeting Band 3 (1800 MHz) for Automotive V2X Communication. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 2657.1 MHz and bandwidth of 276.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Notch Filter
- Frequency Band: Band 3 (1800 MHz)
- Application: Automotive V2X Communication
- Substrate: 128° YX LiNbO3
- Package: LGA (Land Grid Array)
- Design Tool: PathWave EM Design
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Notch Filter design targeting Band 3 (1800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 2657.1 MHz
Bandwidth (3 dB): 276.8 MHz
Insertion Loss Target: <= 1.3 dB
Return Loss Target: >= 21.0 dB
Out-of-Band Rejection: >= 35.0 dB
Isolation (Duplexer): >= 19.0 dB
Q Factor: 10693
Temperature Coefficient:-22.7 ppm/°C
Die Size: 0.7 x 1.2 mm
Wafer Size: 6-inch
Number of Resonators: 4
Electrode Material: W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Notch Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 2657.1 MHz, BW 276.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 7971 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 21.0 dB
- Out-of-band rejection at +/- 415 MHz offset >= 35.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 7.5 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-06-02
Devices Tested: 99
Insertion Loss (meas): 1.03 dB
Return Loss (meas): 23.0 dB
Rejection (meas): 37.9 dB
Isolation (meas): 24.0 dB
Q Factor (meas): 10693
Group Delay Variation: 6.1 ns
Power Handling: +31 dBm
Die Yield: 96.8%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

a) The Notch Filter design demonstrated excellent performance for Band 3 (1800 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. Yuko Ishida	Date: 2024-06-05

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